YEAR 1 · TERM 1 · MODULE 3 · WEEK 1

Lab Workbook

Introduction to Software Engineering Management

Level 1 · 20 min | Level 2 · 30 min | Level 2 · 30 min | Level 3 · 40 min

SE Study Lab · Study material

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Safety rule: run every command only inside ~/se-lab-sandbox or ~/se-lab-remote.git. Read each command before running it.

Level 1 · 20 min

1 Initialise a repository and make your first commit

Goal: Create a repository, check its status, stage a file, and record a commit.

1. Create the sandbox folder

```
mkdir -p ~/se-lab-sandbox && cd ~/se-lab-sandbox
git init -b main
git config user.name "Lab Student"
git config user.email "student@example.com"
```

Expected result: A message says the empty repository was initialised with main as the branch.

Reflect: What changed after this step, and why?

2. Create a file and check status

```
echo "# Sandbox" > README.md
git status
```

Expected result: README.md is listed under "Untracked files".

Reflect: What changed after this step, and why?

3. Stage and commit

```
git add README.md
git status
git commit -m "add README"
git log --oneline
```

Expected result: One commit appears with the message "add README". The status is clean afterwards.

Reflect: What changed after this step, and why?

Level 2 · 30 min

2 Read history and work on a branch

Goal: Inspect history, create a feature branch, commit on it, and merge it back.

1. Create a feature branch

```
cd ~/se-lab-sandbox
git switch -c feature-sum
git branch
```

Expected result: The current branch, marked with *, is feature-sum.

Reflect: What changed after this step, and why?

2. Commit a small feature

```
printf "def add(a, b):\n    return a + b\n" > calc.py
git add calc.py
git commit -m "add sum functionality"
```

Expected result: The commit appears only on feature-sum. main does not contain calc.py yet.

Reflect: What changed after this step, and why?

3. Merge into main

```
git switch main
git merge feature-sum
git log --oneline --graph --all
```

Expected result: A fast-forward merge is reported. The graph shows main and feature-sum at the same commit.

Reflect: What changed after this step, and why?

3 Level 2 · 30 min
Resolve a merge conflict

Goal: Create two conflicting edits on purpose, resolve them, and commit the result.

1. Create the conflict

```
cd ~/se-lab-sandbox && git switch main
echo "Version from main" > greeting.txt && git add greeting.txt && git commit -m "add
greeting from main"
git switch -c feature-greeting HEAD~1
echo "Version from feature" > greeting.txt && git add greeting.txt && git commit -m "add
greeting from feature"
git switch main
git merge feature-greeting || true
```

Expected result: Git reports "CONFLICT (add/add)" and greeting.txt contains <<<<<<< markers.

Reflect: What changed after this step, and why?

2. Resolve and commit

```
printf "Final greeting\n" > greeting.txt
git add greeting.txt
git commit -m "resolve greeting conflict"
git status
```

Expected result: The merge completes. Status is clean and the markers are gone.

Reflect: What changed after this step, and why?

Level 3 · 40 min

4 Connect to a remote and simulate a review

Goal: Use a local bare repository as "origin", push a branch, and practise the fetch and pull difference. Real GitHub steps are listed afterwards.

1. Create a local bare remote

```
cd ~ && git init --bare se-lab-remote.git
cd ~/se-lab-sandbox && git remote add origin ~/se-lab-remote.git
git push -u origin main
```

Expected result: main is uploaded and tracks origin/main.

Reflect: What changed after this step, and why?

2. Push a feature branch

```
cd ~/se-lab-sandbox && git switch -c feature-docs
echo "Docs for sum" > DOCS.md && git add DOCS.md && git commit -m "add docs for sum"
git push -u origin feature-docs
git branch -r
```

Expected result: origin/feature-docs appears in the remote branch list. In GitHub, you would open a pull request from this branch.

Reflect: What changed after this step, and why?

3. Fetch versus pull

```
git fetch origin
git status
git pull origin main
```

Expected result: fetch updates origin/* references; pull integrates origin/main into the current branch.

Reflect: What changed after this step, and why?

Troubleshooting

Symptom	Likely cause	Fix
Author identity unknown	Name or email not configured	git config --global user.name / user.email
Conflict markers in a file	A merge stopped	Edit, remove markers, git add, git commit
Push rejected (non-fast-forward)	The remote has commits you lack	git fetch, integrate, push again. Never --force on shared branches.